
Understanding Sentiments and Topics in News Coverage of Video-Based Social Media and Youth Mental Health – Multi-Label Topic Classification.

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1 Introduction

This project will investigate how news media discuss youth mental-health issues in connection with video-based social-media platforms such as YouTube and Snapchat. The goal will be to automatically identify multiple thematic areas, such as anxiety, depression, cyberbullying, or therapy, from news articles collected by keyword search.

The task will be useful for public-health monitoring, media framing analysis, and computational social-science research, helping researchers understand which mental-health concerns are most emphasized and how coverage differs across platforms or time periods.

The challenge lies in the multi-label nature of the data (each article may address several overlapping topics) and in the noisy retrieval process, since many articles mention mental health or social media only briefly. Additional difficulty arises from limited labeled data and highly imbalanced topic frequencies, requiring careful preprocessing, annotation, and modeling.

2 Problem Formulation

This project will aim to automatically identify the main themes discussed in a collection of news articles. Each article will serve as the input, and the output will be one or more topic categories that describe what the article is about.

Since a single article can belong to more than one category, the task will be treated as a multi-label text classification problem. The model will learn patterns in the text to decide which topics apply to each article.

In short:

- **Input:** Text of a news article.
- **Output:** One or more topic labels.
- **Task Type:** Multi-label classification.

3 Methods

3.1 Data Preprocessing

The project will begin by preparing the text data for analysis. The news articles will first be cleaned by removing unnecessary symbols, numbers, and punctuation. The text will then be converted to lowercase and tokenized into words. Stopwords, or very common words that do not add meaning, will be removed to make the text easier to analyze. This step will ensure that the model focuses on important words and patterns rather than irrelevant details.

3.1.1 Model Design

A text classification model will be used to identify the topics discussed in each article. The model will take the processed text as input and predict one or more topic labels as output. Several model types may be tested, starting with simple baseline methods such as logistic regression or support vector machines, and later moving to more advanced models such as transformer-based architectures. The final choice will depend on performance and available resources.

3.1.2 Loss Function

The model will be trained using a loss function that measures the difference between the predicted labels and the true labels. The goal will be to minimize this difference during training so that the model learns to make accurate predictions. A common choice for this type of task will be binary cross-entropy, which works well when each sample can belong to multiple categories.

3.1.3 Training and Inference

During training, the model will repeatedly see examples of articles and their correct labels. It will adjust its internal parameters to improve prediction accuracy over time. Once training is complete, the model will be evaluated on unseen test data to check how well it can predict topics for new articles. In the inference stage, new articles can be given to the model, and it will output the set of topics it believes are most relevant.

4 Dataset and Experiments

4.1 Dataset Description

The dataset will consist of news articles collected using keyword searches related to social media use and youth mental health. Each article will include a headline and body text. The data will be reviewed and filtered to ensure that only relevant articles are kept. After filtering, each article will be annotated with one or more topic labels that describe the main themes discussed.

Basic statistics such as the total number of articles, average text length, and label distribution will be recorded to understand the dataset characteristics. The data will then be divided into three parts: a training set for model learning, a validation set for tuning model parameters, and a test set for final evaluation.

4.2 Experimental Setup

The experiments will be run using the Python programming language and commonly used NLP libraries such as scikit-learn, PyTorch, and Transformers. Model parameters such as learning rate, batch size, and number of epochs will be adjusted based on validation performance. Training will be performed using an optimizer such as Adam, and early stopping will be applied to prevent overfitting.

4.3 Evaluation

Model performance will be evaluated using standard metrics for multi-label classification, including Precision, Recall, F1-score, and Hamming Loss. The results from both the baseline and advanced models will be compared to see which performs better. Training and validation loss curves will be plotted to show how the model's performance improves over epochs. The best-performing model will then be tested on the held-out test set to assess its generalization ability.

4.4 Expected Outcome

It is expected that the advanced language models will outperform the simpler baseline methods by better capturing contextual meanings in the news text. The model should be able to identify multiple relevant topics per article with reasonable accuracy, helping to uncover how different aspects of youth mental health are discussed in social-media-related news coverage.

5 Project Management

5.1 Team Members

Name	Role	Responsibilities
Ali Abdullah Ahmad	Project Lead & Technical Specialist	Will oversee the entire project workflow, including data filtering, model selection, training, and evaluation. Responsible for designing LLM-assisted prompts for article filtering, implementing multi-label classification models, and analyzing experimental results.
Rishab Apkari	Data Annotator & Evaluation Analyst	Will manage manual and collaborative annotation of news articles, ensuring labeling quality and agreement. Will also assist in calculating inter-annotator agreement (Cohen’s Kappa), preparing evaluation metrics, and contributing to the final report writing.

Table 1: Team members, roles, and responsibilities.

Both team members will collaborate closely on designing the methodology, model selection, and analysis.

5.2 Project Timeline

Phase	Week / Date Range	Key Tasks and Milestones
Project Setup	Oct 6 – Oct 12	Data Preprocessing, finalize topic list, and review the collected dataset.
Step 1: Filtering News Articles	Oct 13 – Oct 19	Perform manual and LLM-assisted filtering to retain only relevant articles related to video-based social media and youth mental health. Finalize the clean dataset for annotation.
Step 2: Annotating the Data	Oct 20 – Oct 26	Conduct manual annotation for topic labels. Use LLM-assisted labeling as initial guidance and validate through human review. Compute inter-annotator agreement using Cohen’s Kappa.
Model Development & Training (Phase 1)	Oct 27 – Nov 2	Implement baseline model (TF-IDF + Logistic Regression) and perform preliminary testing.
Midterm Review Due	Nov 3 Week	Submit midterm review report summarizing progress on data filtering, annotation, and baseline model results.
Midterm Exam Week (No Work)	Nov 10 – Nov 17	Pause all project activities to focus on midterm exams. No major project tasks will be conducted during this period.
Model Development & Training (Phase 2)	Nov 18 – Nov 23	Resume work after midterms. Fine-tune transformer-based model, optimize parameters using validation data, and prepare evaluation results.
Documentation & Final Report Writing	Nov 24 Week	Compile results, finalize figures and tables, and complete the written report for submission.
Presentation & Submission	Dec 1 or Dec 8 Week	Deliver final project presentation and submit report, source code, and supplementary materials.

Table 2: Project timeline with weekly schedule, milestones, and deadlines.